

CS 2614: Computer Organization
Solutions to Homework # 1

Textbook problems

1.

- 3-4. Convert the following decimal numbers to the bases indicated.
- a. 7562 to octal
 - b. 1938 to hexadecimal
 - c. 175 to binary

Solution:

$$\begin{aligned}(7562)_{10} &= (16612)_8 \\ (1938)_{10} &= (792)_{16} \\ (175)_{10} &= (10101111)_2\end{aligned}$$

2.

- 3-5. Convert the hexadecimal number F3A7C2 to binary and octal.

Solution:

$$\begin{aligned}(F3A7C2)_{16} &= (1111\ 0011\ 1010\ 0111\ 1100\ 0010)_2 \\ &= (74723702)_8\end{aligned}$$

Base numbers/Conversion

3. Convert the following binary numbers to decimal:
- a. 1 1010
 - b. 110 1011
 - c. 1101 0110
 - d. 110 0110 1110

Solution:

a.

$$\begin{array}{cccccc} 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 0 & 1 & 0 \end{array} = 2^4 \times 1 + 2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 0$$
$$= 16 + 8 + 0 + 2 + 0$$
$$= 26$$

b.

$$\begin{array}{cccccc} 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \end{array}$$
$$= 2^6 + 2^5 + 2^3 + 2^1 + 2^0$$
$$= 64 + 32 + 8 + 2 + 1$$
$$= (107)_{10}$$

c.

$$\begin{array}{cccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \end{array}$$
$$= 2^7 + 2^6 + 2^4 + 2^2 + 2^1$$
$$= 128 + 64 + 16 + 4 + 2$$
$$= (214)_{10}$$

d.

$$\begin{array}{ccccccc} 2^{10} & 2^9 & & 2^6 & 2^5 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{array}$$
$$= 2^{10} + 2^9 + 2^6 + 2^5 + 2^3 + 2^2 + 2^1$$
$$= 1024 + 512 + 64 + 32 + 8 + 4 + 2$$
$$= 1646$$

4. Convert the following decimal numbers to binary:

- a. 81
- b. 181

Solution:

a.

81	
40	
20	0
10	0
5	0
2	1
1	0
0	1

$$(81)_{10} = (1010001)_2$$

181	
90	1
45	0
22	1
11	0
5	1
2	1
1	0
0	1

$$= (10110101)_2$$

5. Convert the following binary numbers to hexadecimal:

- a. 10100111011
b. 11110001110
c. 10011110011101
d. 10100101011111100001010

Solution:

a.

$$(10100111011)_2 = (53B)_{16}$$

b.

$$\underbrace{1111}_{7} \underbrace{000}_{8} \underbrace{1110}_{E} = (78E)_{16}$$

c.

$$\begin{array}{cccc} 1 & 001 & 111 & 0011101 \\ \hline 2 & 7 & 9 & D \end{array} \quad (279D)_{16}$$

d.

$$\begin{array}{cccccc} 10 & 1001 & 0101 & 1111 & 0000 & 1010 \\ \hline 2 & 9 & 5 & F & 0 & A \end{array}$$

$$= 295FOA$$

6. Convert the following hexadecimal numbers to binary:

- a. D9
- b. 483
- c. 80BF
- d. BA0E5

Solution:

a.

$$(D9)_{16} = \begin{array}{cc} (1101 & 1001) \\ \hline D & 9 \end{array} 2$$

b.

$$483 = \begin{array}{ccc} 100 & 1000 & 011 \\ \hline 4 & 8 & 3 \end{array}$$

c.

$$80BF = \frac{1000}{8} \frac{0000}{0} \frac{1011}{B} \frac{1111}{F}$$

d.

$$BA0E5 = \frac{1011}{B} \frac{1010}{A} \frac{0000}{0} \frac{1110}{E} \frac{0101}{5}$$

7. Convert the following numbers to the bases indicated:

From	To
$(110011)_2$	Decimal
$(1011.101)_2$	Decimal
$(10110101)_2$	Octal
$(10101011101)_2$	Hexadecimal
$(51)_{10}$	Binary
$(177)_{10}$	Octal
$(77)_{10}$	Hexadecimal
$(632)_8$	Decimal
$(F4C)_{16}$	Decimal
$(1101101.1101)_2$	Octal
$(1101101.1101)_2$	Hexadecimal

Solution:

$$\begin{array}{r} 2 \ 2 \ 0 \ 0 \ 2 \ 2 \\ = 2^0 + 2^1 + 2^4 + 2^5 = 2 + 2 + 26 + 32 = \\ 52_{10} \end{array}$$

$$\begin{array}{cccc} 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1} & 2^{-2} & 2^{-3} \\ (1 \ 0 \ 11.101)_2 \end{array}$$

$$\begin{aligned} &= (2^3 + 2^1 + 2^0) \cdot (2^{-1} + 2^{-3}) \\ &= (8 + 2 + 1) \cdot (0.5 + 0.125) \\ &= (11.625)_{10} \end{aligned}$$

Conversion of binary to octal (base 2 to base 8):

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$$\begin{array}{|c|c|c|} \hline 10 & 110 & 101 \\ \hline 2 & 6 & 5 \\ \hline \end{array} = (265)_8$$

Conversion of binary to hexadecimal (base 2 to base 16):

Hexadecimal	Binary
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

$$\begin{array}{|c|c|c|c|} \hline 1010 & 1011 & 1011 \\ \hline 5 & 5 & D \\ \hline \end{array} = (55D)_{16}$$

The rules for conversion from binary to hexadecimal are as given below:

1. Divide the given binary number into a group of four bits from binary point to left (or from right to left if no fractional part) and from binary point to the right. Append 0's at leading or trailing position if necessary to make each group of 4 bits.
2. Substitute each group of four bits by hexadecimal equivalent symbol (letter or digit) from the table.
3. Collect Hexadecimal symbols to get Hexadecimal equivalent number

$$\begin{array}{r|l}
 (51)_{10} & \\
 25 & 1 \\
 12 & 1 \\
 6 & 0 \\
 3 & 0 \\
 1 & 1 \\
 0 & 1
 \end{array} = (110011)_2$$

$$\begin{array}{r|l}
 (177)_{10} & \\
 22 & 1 \\
 2 & 6 \\
 0 & 2
 \end{array} = (261)_8$$

$$\begin{array}{r|l}
 (77)_{10} & \\
 4 & (13)_{10} \\
 \textcircled{1} & 4
 \end{array} = (4D)_{16}$$

$$\begin{aligned}
 (632)_8 &= 6 \times 8^2 + 3 \times 8^1 + 2 \times 8^0 \\
 &= 384 + 24 + 2 \\
 &= (410)_{10}
 \end{aligned}$$

$$(F4C)_{16} = F \times 16^2 + 4 \times 16^1 + C \times 16^0$$

$$= 15 \times 256 + 4 \times 16 + 12 \times 1$$

$$= 3840 + 64 + 12$$

$$= 3916$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 & 1 & 0 & 1 \end{array} \right)_8 \quad \left[\begin{array}{c} \cdot 1101 = \cdot 1101 \end{array} \right]$$

$$= (155.64)_8$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 & 1 & 0 & 1 \end{array} \right)_{16}$$

$$= 6D.D$$